C PROGRAMMING

Lesson 6 — Useful Math Functions

Based on: Bro Code C Tutorial Series

1. The math.h Header

To use any math function in C, you must include the math.h library at the top of your file. Without it, the compiler will throw warnings and the functions may not work correctly.

```
#include <stdio.h>
#include <math.h>   // required for all math functions
```

> **Rule:** Always add #include <math.h> whenever you use sqrt, pow, round, ceil, floor, log, sin, cos, or tan.

2. Square Root — sqrt()

Returns the square root of a number. The result is always a double.

```
double result = sqrt(9);      // result = 3.0
```

3. Power — pow()

Raises a base number to the power of an exponent. Both arguments and the return value are doubles.

```
printf("%lf\n", pow(2, 4));   // Output: 16.000000 (2^4)
```

4. Rounding Functions

C has three different rounding functions depending on the direction you want to round:

round() — nearest integer

```
printf("%lf\n", round(3.14));  // Output: 3.0  (rounds down)
printf("%lf\n", round(3.5));   // Output: 4.0  (rounds up at .5)
```

ceil() — always round UP

```
printf("%lf\n", ceil(3.1));    // Output: 4.0
```

floor() — always round DOWN

```
printf("%lf\n", floor(3.9));   // Output: 3.0
```

5. Absolute Value — abs()

Returns the positive (non-negative) version of a number. Think of it as the distance from zero — distance is never negative.

```
#include <stdlib.h>   // abs() for integers is in stdlib.h

printf("%d\n", abs(-3));    // Output: 3
```

> **Note:** abs() works on integers. For doubles, use fabs() from math.h instead.

6. Natural Logarithm — log()

Returns the natural logarithm (base e) of a number. Mainly used in scientific and mathematical calculations.

```
printf("%lf\n", log(3));    // Output: 1.098612
```

7. Trigonometric Functions

C provides sin(), cos(), and tan() for trigonometry. All of them take input in RADIANS, not degrees.

```
// Input must be in radians!
// To convert degrees to radians:
// radians = degrees * (3.14159 / 180)
double angle = 45 * (3.14159 / 180);   // 45 degrees in radians
printf("%lf\n", sin(angle));            // Output: 0.707107
```

> **Important:** sin/cos/tan take RADIANS not degrees. To convert: radians = degrees × (π / 180). This is easy to forget!

8. Practice Problem

> **Practice: Math Calculator**

Write a C program that asks the user to enter a number, then prints the following results for that number:

- Its square root
- Its square (number to the power of 2)
- Its cube (number to the power of 3)
- Its absolute value (enter a negative number to test this)
- Rounded down and rounded up values (use a decimal number)

Expected Output (for input: -7.6)

```
Enter a number: -7.6

Square root of absolute value: 2.757
Squared:  57.76
Cubed:    -438.976
Absolute value: 7.6
Floor: -8.0
Ceil:  -7.0
```